

with me to translate and adapt the communication boards into Malayalam. This collaborative effort showed how community participation can make open source assistive technologies more accessible and relevant to local users. However, this experience highlighted a larger issue.

Even today, support for Indian languages remains limited. Languages spoken by millions of people, including Tamil, Telugu, Kannada, Marathi, Gujarati and Odiya, need stronger representation in assistive communication technologies.

Language support involves much more than translation. The quality and naturalness of speech output depend not only on the AAC software but also on the underlying text-to-speech (TTS) engine available for the selected language. Pronunciation is often inaccurate; speech output may sound anglicised and automated translations can produce poor results. These issues affect the overall usability of AAC systems.

Why localisation matters

Many communication boards are designed for users in the West. For example, consider communication boards representing various food items, where the pictures and vocabulary should reflect foods that are used daily. These often contain items such as pancakes, pasta or porridge.

When we worked with Malayalam-speaking users, we found it more useful to create customised boards containing familiar items such as rice, sambar, papadam and water. A communication tool becomes more effective when it reflects the language and culture of the people using it.

Beyond software

Software alone is not enough. During our work, we tested the software on a Lenovo IdeaPad Flex 5 convertible laptop powered by an AMD Ryzen 5 7530U processor with 16GB RAM

and a 35.56 cm (14-inch) touchscreen display. For audio output, we used a Mivi Play portable Bluetooth speaker, which provided clear and sufficiently loud speech synthesis during demonstrations.

Based on our experience, a touchscreen laptop of this kind is the most recommended hardware for AAC applications because the larger interactive display and full computing capabilities make it easier for many users to communicate effectively. However, specialised hardware is not essential. CBoard also runs well on tablets and even smartphones, making assistive communication accessible to individuals and institutions with limited resources.

We found that touchscreen laptops often work better for many users than smartphones or tablets. Some individuals have difficulty holding devices, while others require larger interfaces for interaction. The customised software and the touchscreen laptop used during testing were later donated to a special school, demonstrating that the project was not merely a proof-of-concept but a practical solution deployed for real users.

Screen readers are also important because many users cannot read text independently. Gaze-tracking systems can help individuals who are unable to use conventional input methods. These technologies can significantly improve communication and independence.

There is also a need for affordable open source hardware solutions. Refreshable Braille displays and accessibility devices remain expensive for many users and institutions.

The role of community participation

One lesson I learned is that accessibility cannot be achieved through technology alone. At the same time, accessibility solutions should be developed in accordance with

recognised accessibility standards and universal design principles so that they are usable by the widest possible range of people.

Community participation is essential. Automated translation often produces errors, and many words require local knowledge to be translated correctly. Communities can contribute by creating vocabulary sets, correcting translations and developing culturally relevant communication boards.

India also needs better image datasets and audio datasets for regional languages. As AI-powered tools become more common, the availability of quality language resources will become increasingly important.

It was wonderful to see initiatives where communities contribute language recordings and datasets. Such efforts can play an important role in improving accessibility technologies in the future.

Looking ahead

I would like to end with an example that I often think about. María del Mar Galcerán Gadea, a politician from Spain who has Down syndrome, was elected to the Valencian parliament. Her story shows that people can overcome significant challenges when society provides the right opportunities and support.

The same principle applies to communication accessibility. Technology alone cannot solve every problem, but it can remove many barriers. Open source platforms such as CBoard, combined with community participation and stronger support for Indian languages, can help create more inclusive communication systems. If we continue working in this direction, we can ensure that more people have the opportunity to express themselves, participate in society and live with dignity. **END** 🐼

This article is based on a talk by Dr Deepu Benson, a free software enthusiast, at Open Source India 2025. Transcribed and curated by Gayathri M. from the EFY Group.